

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Sideband Register Type Change
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:
Changing the type of the fields in Sideband Register 13 to be Read-Only

Benefits as a result of the changes:
These fields are not required to be Writeable.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
Need to verify the new type in Compliance.

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Actual Change

(a). Table 4-20 – SB Register Fields

To Text:

13	Gen 2/3 TxFFE	0	3:0	TxFFE Request (Lane 0) – Identifies one of 16 predefined TxFFE configurations requested by the receiver.	RO	Vendor Defined
			4	Rx Locked (Lane 0) – Indicates that the receiver completed TxFFE negotiation. 0b – TxFFE negotiation is not done 1b – TxFFE negotiation is done	RO	0b
			5	Rx Active (Lane 0) – Indicates whether the receiver is active. 0b – Receiver is inactive 1b – Receiver is active	RO	0b
			6	Clock Switch Done (Lane 0) Shall be set to the value of the <i>Rx Locked (Lane 0)</i> bit.	RO	0b
			7	New Request (Lane 0) – Indicates whether the receiver is processing a TxFFE Request. 0b – Receiver finished processing the previous request 1b – Receiver is still processing the previous request	RO	0b
		1	3:0	TxFFE Request (Lane 1) – Identifies one of 16 predefined TxFFE configurations requested by the receiver.	RO	Vendor Defined
			4	Rx Locked (Lane 1) – Indicates that the receiver completed TxFFE negotiation. 0b – TxFFE negotiation is not done 1b – TxFFE negotiation is done	RO	0b
			5	Rx Active (Lane 1) – Indicates whether the receiver is active. 0b – Receiver is inactive 1b – Receiver is active	RO	0b
			6	Clock Switch Done (Lane 1) Shall be set to the value of the <i>Rx Locked (Lane 1)</i> bit.	RO	0b
			7	New Request (Lane 1) – Indicates whether the receiver is processing a TxFFE Request. 0b – Receiver finished processing the previous request 1b – Receiver is still processing the previous request	RO	0b
		2	3:0	TxFFE Setting (Lane 0) – Index of the TxFFE configuration to be loaded to the transmitter.	ROW	Vendor Defined
			5:4	Reserved	Rsvd	0

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

			6	Request Done (Lane 0) – Indicates whether the transmitter loaded the recent requested index of TxFFE configuration. 0b – Transmitter has not yet loaded the recent requested index of TxFFE configuration 1b – Transmitter has loaded the recent requested index of TxFFE configuration	ROW	0b
			7	Tx Active (Lane 0) – Indicates whether the transmitter is transmitting a valid signal. 0b – Transmitter is not transmitting a valid signal 1b – Transmitter is transmitting a valid signal	ROW	0b
		3	3:0	TxFFE Setting (Lane 1) – Index of the TxFFE configuration to be loaded to the transmitter.	ROW	Vendor Defined
			5:4	Reserved	Rsvd	0
			6	Request Done (Lane 1) – Indicates whether the transmitter loaded the recent requested index of TxFFE configuration. 0b – Transmitter has not yet loaded the recent requested index of TxFFE configuration 1b – Transmitter has loaded the recent requested index of TxFFE configuration	ROW	0b
			7	Tx Active (Lane 1) – Indicates whether the transmitter is transmitting a valid signal. 0b – Transmitter is not transmitting a valid signal 1b – Transmitter is transmitting a valid signal	ROW	0b